

THE ELEMENTS OF TEACHING AND LEARNING

Focus Questions:

- What is the nature of the learner?
- With what is the learner equipped that enables him/her to learn?
- What professional and personal qualities must a teacher possess to facilitate learning?
- What makes a conducive learning environment?
- What are the implications of these thoughts on the learner, the teacher, and learning environment to teaching-learning?

Introduction

The principal elements that make teaching and learning possible and attainable are the teacher, the learner, and a conducive learning environment. Only when a positive relationship exists among them can teaching and learning occur with precision and predictability.

The teacher serves as the prime mover of the educational wheel while the learner is the key participant in the learning process. The favorable environment provides essential features and ingredients that could make a headway in guiding the teaching-learning processes and methodologies needed for a smooth linkage among the three.

What role does each play in the teaching-learning process? How does learning environment contribute to learning?

The learner and the teacher are the key players in a learning environment.

The Learner

"Every child is a potential genius."

Focus Questions:

- What is the nature of the learner? What are the implications of these to the teaching-learning process?
- What are the powers/faculties with which every learner is equipped?
- What should teachers do and not do to help develop the learner's powers or faculties?
- What does Howard Gardner say about the learner's intelligence?
- What are learning styles? How do these affect teaching-learning?

The Learner as an embodied spirit

The learner is an embodied spirit. He is neither body nor spirit alone.

The learner has the power to see, hear, touch, smell and taste, perceive, imagine, retain, recall, recognize past mental acts, conceive ideas, make judgment, reason out, feel and choose.

The learner is an **embodied spirit**. He/she is a union of a sentient body and a rational soul. His/her body experiences sensations and feels pleasure and pain. His/her soul is the principle of spiritual acts, the source of intellectual abstraction, self-reflection, and free rational volition. Body and soul exist in mutual dependence. (Kelly, 1965) As teachers then, let us care for the embodied spirit-learner. Let us feed his/her body as well as his/her spirit.

A learner who is hungry, physically exhausted and sleepy cannot be at his/her best in the classroom. That's why a school cannot ignore the learner's physical needs. There are a lot of practices in school that show that the school cares for his/her physical needs. We hear or read about feeding programs in schools. The teacher is helped by a team of health personnel that visits every school to check on the health condition of learners. The team measures the learners' height, take their weight, check their teeth and gums, eyesight, and heads for lice etc. There is period for recess, for brushing teeth, for necessity. Teacher inserts "energizers" such as action songs, short physical exercise, or just a song.

The learner has also a spiritual nature. So he/she must nourish not only his body but also his spirit. "Man does not leave by

bread alone but by every word that comes from the mouth of God”, said the Greatest Teacher. The learner needs “chicken soup for the soul” or else suffers from “spiritual vitamin deficiency”. He/she needs to be fed with sublime thoughts, words of inspiration, spiritual advice drawn from the Bible for Christians, Koran for Muslims, the Vedas for the Hindus, Gautama’s Buddha’s teachings for the Buddhists.

Only when the learner is nourished materially and spiritually can he/she be complete for that is his/her very nature.

The Fundamental Equipment of the Learner

Cognitive Faculties

With which faculties or powers is the learner equipped so he/she can learn? He/She is equipped with **cognitive** as well as appetitive **faculties**. His/her cognitive faculties include the following: 1) five senses, 2) instinct, 3) imagination, 3) memory, and 4) intellect. By his/her senses, the learner is able to see, hear, feel, taste and smell whatever is to be learned. By the power of imagination, the learner is able to form representations of material objects which are not actually present to the senses. By his/her power of memory he is able to retain, recall and recognize past mental acts. By his/her intellect, s/he can form concepts or ideas, makes judgment, and reason out.

Five senses. The five (5) senses are part of the learner’s sentient body. For effective and efficient learning, it is important that his/her senses function normally. It is said that “*there is nothing in the mind which was not first in some manner in the senses.*” Do you agree? Which of the senses contribute most to learning? What is the contribution of each of the senses to learning? (See Figure 4 in Unit III, Chapter 3 for the answer.)

Instincts. The learner is also equipped with instincts. The word instinct comes from the Latin word *instinctus* which means impulse. This means that the learner has a natural or inherent capacity or tendency to respond to environmental stimuli such as danger signs for survival or self-preservation. This is manifested in his/her immediate tendency to flee in case of danger or to fight when attacked or to rationalize to defend himself/herself when his/her ego is hurt.

For the learner to learn and develop into a human person, the teacher must teach and the learner to put his/her instincts under control. If not, he/she will not be different from any brute that is bound by its instinct and will be far from becoming the human person who is capable of understanding, reasoning, choice and self-control that he/she is meant to become.

Imagination. This is another cognitive faculty of the learner. It is the ability to form a mental image of something that is not perceived through the senses. It is the ability of the mind to build mental scenes, objects or events that do not exist, are not present or have happened in the past.

There are many instances in the teaching-learning process that call for the learner's power of imagination. His/her imaginative power is challenged in role playing, simulation, short story writing, drawing, dress designing, visualizing a situation and in inventing creative ways of reporting or presenting. The teaching-learning process will be bare and dry without the use of imagination. The learner's creative powers cannot be unleashed without the use of imagination.

The teacher must, therefore, help the learner develop his/her power of imagination by encouraging them to "think outside the box", to be creative, to form new ideas and explore old ideas.

Memory. This is the cognitive faculty of retaining and recalling past experience. As the teaching-learning process goes on, you will have a lot to commit to memory-formula in finding the area of a circle, the stages of mitosis, striking passages from a literary piece, presidents of the Philippines, grammar rules and the like. Memory work is basic in learning but, of course, we do not encourage memorizing without understanding. The learner must, therefore, be helped to commit things to memory. How? One way is by making him/her repeat the information mentally again and again such as repeating the cell phone number you want him/her to commit to memory. Another way is by associating the information you want him/her to remember with something he/she already knows. The learner remembers Rizal's birthday, June 19 because it also his/her birthday.

A teacher will do the learner a favor when he/she does not bombard the learners with too much information too rapidly and when he/she allocates time for rehearsal / verbal repetition of lessons during classroom lessons. When he/she pauses once in a while to ask learners whether they have any questions, he/she gives the learners a few moments to think over and mentally rehearse

what they have just learned.

Intellect. This is another cognitive faculty of the learner. By his/her intellect, the learner can engage in cognitive processes such as forming ideas or concepts, reasoning out and making judgment. A child and his/her mother sees a black dog. The mother tells him/her "this is a dog" and so the child forms a concept of a dog. It has four legs, two eyes, black, covered with hair. The next day, they go to a neighbor's house and see another breed of dog. The mother tells the child, "this is another dog." The child's concept of dog expands. Dogs come in different colors and size. This process of concept formation is possible because of the child's intellect. Early concepts get modified and expanded as the child grows and develops. There is so much concept formation that takes in every teaching-learning process.

The same intellect enables the learner to reason out and judge. The use of syllogism in logic illustrates the 3 cognitive processes of conception or concept formation, reasoning and judging. Here is an example: *All men are rational.*

Pedro is a man.

Therefore, Pedro is rational.

There are two concepts introduced, "man/men" and "rational". Relating the concepts and seeing the consistency of the relation of the concepts to each other is the essence of logical reasoning. This reasoning leads to judgment, the conclusion, "Therefore, Pedro is rational."

Reasoning includes analyzing. Judging is evaluating. Notice that analyzing and evaluating are in Bloom's cognitive taxonomy of objectives. Other cognitive processes are classifying, inferring, drawing generalizations, synthesizing. You add to the list when you think of 21st century skills.

Appetitive Faculties

The learner's appetitive faculties are 1) his/her feelings and emotions and 2) rational will.

Feelings and emotions. Emotion is the on / off switch for learning. Positive feelings and emotions make the teaching-learning process an exciting and a joyful, fruitful affair. Negative feelings and emotions make the same process a burden. The lessons that we learn and remember most are those that have struck us in one way or another.

Learners differ in their abilities, aptitudes, interests, home background values and attitudes.

The differences among learners become more accentuated with the integration of children with special needs and children from the indigenous peoples (IP) group in the classroom.

Negative emotions adversely affect the cognitive processes of recalling, imagining, analyzing, reasoning, judging, evaluating synthesizing. Faced with frustration, despair, worry, sadness, or shame, learners lose access to their own memory, reasoning, and the capacity to make connections.

The mere thought of being asked to read aloud in class is enough to freeze some learners. Having to take a written test or oral exam, which require memory, reasoning, classifying, synthesizing can lock some learners' gears. The sight of a math word problem knocks down some learners. You scare learners and they perform poorly and don't learn new information well. Anxiety is the enemy of memory. Unfortunately, in many of today's classrooms, we see learners whose intellectual energies and capacities are drained by negative emotional states.

All learners are equipped with the cognitive and appetitive faculties. They differ however in the degree to which they are utilized and expressed on account of the learners' abilities, aptitudes, interests, values and attitudes and home background. Let us take a look once more at the learner from the point of view of these five distinguishing elements.

Will. The learner's will serves as guiding force and the main integrating force in his/her character. By his/her will, the learner wills what his/her intellect presents as good and desirable. It is this will that makes the learner free to choose or not to choose to do the good as presented by his/her intellect. It is this free will that will not allow the learner to be totally determined by his/her environment. This means that the degree to which the learner is influenced by his/her environment depends ultimately the strength of his/her will. The learner whose will is weak will easily succumb to the bad influence of his/her peer group even if his/her intellect tells him/her not. But the learner with a strong will shall resist the temptation to be influenced by bad peer group. Therefore, the focus of values education should be the strengthening of the will.

Factors that contribute to the differences among learners.

All learners are equipped with the cognitive and appetitive faculties. They differ however in the degree to which they are utilized and expressed on account of the learners' abilities, aptitudes, interests, values and attitudes and home background. Let us take a look once more at the learner from the point of view of these five distinguishing elements.

1. Ability

The learners' native ability dictates the prospects of success in any purposeful activity. Hence, the learners' proficiency in memorization, imagination concept formation, reasoning, judging and other cognitive skills are contingent on their endowed potential to learn. Ability determines the learners' capacity to understand and assimilate information for their

own use and application. As learners they differ in the way they observe and interpret happenings in their surroundings. Some are more perceptive and discerning while others are less inquisitive. With such typical reactions and facility to learn, they may be classified generally into fast, average and slow learners. Others are labeled high, moderate and slow achievers. As to their mental ability, students can be categorized into superior, above average, average and below average. A wide range in their intelligence is a factor to consider in planning instruction.

2. Aptitude

Aptitude refers to the learners' innate talent or gift. It indicates a natural capacity to learn certain skills. The powers of memory, imagination, concept formation, reasoning and judgment on matters related to the arts function best for those who exhibit special inclination for the arts such as painting and designing crafts, propensity for music and flair for dramatics. Likewise, the same cognitive powers are at their peak for mathematics for those with aptitude in math.

An early recognition of said natural adeptness among learners is indeed compelling so as not to waste such aptitude. Provisions of a formative environment will be of great help in enabling them to flourish and grow.

3. Interests

Learners' interest in learning makes learning no longer a task but a pleasure. The learners' cognitive faculties of sensorial experience, memory, imagination, concept formation, reasoning and judgment are at their height when learners' interests are also at its peak.

Learners have varied interests. A physically robust student would go for athletics, while an artistic and stylish student would pursue hobbies that are fascinating. Girls are strongly attracted to flowering plants and greeneries and their preoccupations revolve around them. Boys go for hiking and mountain climbing. Of course, there will always be exceptions.

Interests are not inherited. They are developed. A classroom set-up could offer centers of interest to give learners an opportunity to develop interests in many things. Interest clubs organized by different disciplines may serve as outlet of special interests shared by the members.

4. Family and cultural background

Students who come from different socioeconomic background manifest a wide range of behavior due to differences in upbringing practices. Some families allow their members to express their preferences regarding self-discipline while others are left to passively follow home regulations. Their participation in classroom activities are influenced by their home training and experiences, either they become attuned and confident in their ways or inactive and apathetic.

Today, we speak of multi-cultural, diverse, pluralistic classrooms.

Beneficial relationships of learners with their mentors and with one another affirm the kind of bond they enjoy at home. Cooperation, coupled with a willingness to share, is instilled and is carried over to all associations they join. The tendency to readily affiliate with a group is most welcomed in a classroom setting where teamwork achieves desired lesson objectives.

5. Attitudes and values

A positive attitude will enhance the maximum and optimum use of the learner's cognitive and affective faculties for learning. A negative attitude towards learning robs them of many opportunities for learning. Learners with a positive attitude will demonstrate the value of persistence in their studies. Persistent students sustain interest in a learning activity not mindful of the extra time and effort being spent. They pursue the task to completion and never give up when confronted with problems. They develop the attitude of trying alternative procedures until they obtain satisfactory results. They are driven by a never-ending search for more knowledge and information.

We will always find time for things we consider to be of value. If we value learning, we will give it a priority. If we don't value it, we will always find a ready excuse for not engaging in it.

When we have positive learning beliefs and attitudes, we can relax, remember, focus and absorb information as we learn.

An Exercise

Use this exercise to explore your attitudes and beliefs about learning. Put a check mark beside the statements that are TRUE of you:

- ☐ 1. There will always be something new to learn for as long as I live.
- ☐ 2. I'm not afraid to commit mistakes as long as I learn from them.
- ☐ 3. I take advantage of every opportunity to learn.
- ☐ 4. I take charge of my own learning.
- ☐ 5. I am very willing and eager to learn.
- ☐ 6. I am grateful to teachers who demand quality and excellence for effective learning.
- ☐ 7. I have made it habit to set my personal learning goals.
- ☐ 8. I find time to check my progress against my personal learning goals.
- ☐ 9. I give my best in everything I am asked to do in order to learn.

How many did you score out of nine (9) highest possible score? Review the statements which you didn't check. How can these unchecked items affect your learning?

For each statement that you didn't check, think of two steps that you could take right now to build a positive learning attitude in this area.

Gardner's Multiple Intelligence Theory

The learner is gifted with intelligence. The concept of intelligence evolved through the years. Intelligence then was measured only in terms of language and mathematical competence. With Howard Gardner's theory on multiple intelligences, the concept of the learner's intelligence has gone beyond linguistic and mathematical intelligences. There are seven more intelligences in addition to linguistic and mathematical intelligences. (Please refer to Figure 1 given on the next page).

FIGURE 1 Intelligences as Dispositions

Disposition/Intelligence	Sensitivity to:	Inclination to:	Ability to:
1. Verbal-Linguistic Intelligence	sounds, meanings, structures and styles of language	speaking, writing, listening, reading	speak effectively (teacher, religious leader, politician) or write effectively (poet, journalist, novelist, copywriter, editor)
2. Logical-Mathematical Intelligence	patterns, numbers and numerical data, causes and effects, objective and quantitative reasoning	finding patterns, making calculations, forming and testing hypothesis, using the scientific method, deductive and inductive reasoning	work effectively with numbers (accountant, statistician, economist) and reason effectively (engineer, scientist, computer programmer)
3. Spatial Intelligence	colors, shapes, visual puzzles, symmetry, lines, images	representing ideas visually, creating mental images, noticing visual details, drawing and sketching	create visually (artist, photographer, engineer, decorator) and visualize accurately (tour guide, scout, ranger)
4. Bodily-Kinesthetic Intelligence	touch, movement, physical self, athleticism	activities requiring strength, speed, flexibility, hand-eye coordination, and balance	use the hands to fix or create (mechanic, surgeon, carpenter, sculptor, mason) and use the body expressively (dancer, athlete, actor)
6. Musical Intelligence	tone, beat, tempo, melody pitch, sound	listening, singing, playing an instrument	create music (song writer, composer, musician conductor) and analyze music (music critic)
7. Interpersonal Intelligence	body language, moods, voice, feelings	noticing and responding to other people's feelings and personalities	work with people (administrators, managers, consultants, teachers) and help people identify and overcome problems (therapists, psychologists)
8. Intrapersonal Intelligence	one's own strengths, weaknesses, goals, and desires	setting goals, assessing personal abilities and liabilities, monitoring one's own thinking	meditate, reflect, exhibit self-discipline, maintain composure, and get the most out of oneself
*Naturalist Intelligence	natural objects, plants, animals, naturally occurring patterns, ecological issues	identifying and classifying living things and natural objects	analyze ecological and natural situations and data (ecologists and rangers), learn from living things (zoologist, botanist, veterinarian) and work in natural settings (hunter, scout)
9. *Existential Intelligence	"big picture" of existence; why are we here	sensitivity to deep questions about human existence, such as the meaning of life why do we die, and how did we get here.	reflect on the meaning of human existence

FIGURE 1

(Source: Harvey F. Silver, et al. *So Each May Learn*, Virginia: Association of Supervision and Curriculum Development 2000.- Gardner's ninth intelligence is not part of the original Table given by Harvey F. Silver.)

* Added by the authors

Find out your multiple intelligences. Take this test.

This form can help you determine which intelligences are strongest for you... Many thanks to Dr. Terry Armstrong for graciously allowing us to use her questionnaire.

Instructions: Read each statement carefully. Choose one of the five buttons for each statement indicating how well that statement describes you.

- 1 = Statement does not describe you at all
- 2 = Statement describes you very little
- 3 = Statement describes you somewhat
- 4 = Statement describes you pretty well
- 5 = Statement describes you exactly

	1	2	3	4	5
1. I pride myself on having a large vocabulary.	☐	☐	☐	☐	☐
2. Using numbers and numerical symbols is easy for me.	☐	☐	☐	☐	☐
3. Music is very important to me in daily life.	☐	☐	☐	☐	☐
4. I always know where I am in relation to my home.	☐	☐	☐	☐	☐
5. I consider myself an athlete.	☐	☐	☐	☐	☐
6. I feel like people of all ages like me.	☐	☐	☐	☐	☐
7. I often look for weaknesses in myself that I see in others.	☐	☐	☐	☐	☐
8. The world of plants and animals is important to me.	☐	☐	☐	☐	☐
9. I enjoy learning new words and do so easily.	☐	☐	☐	☐	☐
10. I often develop equations to describe relationships and/or to explain my observations.	☐	☐	☐	☐	☐
11. I have wide and varied musical interests including both classical and contemporary.	☐	☐	☐	☐	☐
12. I do not get lost easily and can orient myself with either maps or landmarks.	☐	☐	☐	☐	☐
13. I feel really good about being physically fit.	☐	☐	☐	☐	☐
14. I like to be with all different types of people.	☐	☐	☐	☐	☐

	1	2	3	4	5
15. I often think about the influence I have on others.	☐	☐	☐	☐	☐
16. I enjoy my pets.	☐	☐	☐	☐	☐
17. I love to read and do so daily.	☐	☐	☐	☐	☐
18. I often see mathematical ratios in the world around me.	☐	☐	☐	☐	☐
19. I have a very good sense of pitch, tempo, and rhythm.	☐	☐	☐	☐	☐
20. Knowing directions is easy for me.	☐	☐	☐	☐	☐
21. I have good balance and eye-hand coordination and enjoy sports which use a ball.	☐	☐	☐	☐	☐
22. I respond to all people enthusiastically, free of bias or prejudice.	☐	☐	☐	☐	☐
23. I believe that I am responsible for my actions and who I am.	☐	☐	☐	☐	☐
24. I like learning about nature.	☐	☐	☐	☐	☐
25. I enjoy hearing challenging lectures.	☐	☐	☐	☐	☐
26. Math has always been one of my favorite classes.	☐	☐	☐	☐	☐
27. My music education began when I was younger and still continues today.	☐	☐	☐	☐	☐
28. I have the ability to represent what I see by drawing or painting.	☐	☐	☐	☐	☐
29. My outstanding coordination and balance let me excel in high-speed activities.	☐	☐	☐	☐	☐
30. I enjoy new or unique social situations.	☐	☐	☐	☐	☐
31. I try not to waste my time on trivial pursuits.	☐	☐	☐	☐	☐
32. I enjoy caring for my house plants.	☐	☐	☐	☐	☐
33. I like to keep a daily journal of my daily experiences.	☐	☐	☐	☐	☐
34. I like to think about numerical issues and examine statistics.	☐	☐	☐	☐	☐

4 5

1 2 3 4 5

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|--|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| 35. I am good at playing an instrument and singing. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 36. My ability to draw is recognized and complemented by others. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 37. I like being outdoors, enjoy the change in seasons, and look forward to different physical activities each season. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 38. I enjoy complementing others when they have done well. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 39. I often think about the problems in my community, state, and/or world and what I can do to help rectify any of them. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 40. I enjoy hunting and fishing. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 41. I read and enjoy poetry and occasionally write my own. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 42. I seem to understand things around me through a mathematical sense. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 43. I can remember the tune of a song when asked. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 44. I can easily duplicate color, form, shading, and texture in my work. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 45. I like the excitement of personal and team competition. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 46. I am quick to sense in others dishonesty and desire to control me. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 47. I am always totally honest with myself. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 48. I enjoy hiking in natural places. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 49. I talk a lot and enjoy telling stories. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 50. I enjoy doing puzzles. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 51. I take pride in my musical accomplishments. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 52. Seeing things in three dimensions is easy for me, and I like to make things in three dimensions. | ☐ | ☐ | ☐ | ☐ | ☐ |
| 53. I like to move around a lot. | ☐ | ☐ | ☐ | ☐ | ☐ |

	1	2	3	4	5
54. I feel safe when I am with strangers.	☐	☐	☐	☐	☐
55. I enjoy being alone and thinking about my life and myself.	☐	☐	☐	☐	☐
56. I look forward to visiting the zoo.	☐	☐	☐	☐	☐
57. Most often I ponder on the question "Why am I here in this world?"	☐	☐	☐	☐	☐
58. I often reflect on why innocent people suffer?	☐	☐	☐	☐	☐
59. I spend my time reading inspirational books.	☐	☐	☐	☐	☐
60. I want to live my life meaningfully.	☐	☐	☐	☐	☐
61. What's my mission in life?	☐	☐	☐	☐	☐
62. Is there such a thing as hell?	☐	☐	☐	☐	☐
63. Where can I find happiness?	☐	☐	☐	☐	☐

Interpretation

The following items in the questionnaire can be categorized into the nine intelligences:

- | | | |
|-------------------|---|----------------------------|
| 1. Linguistic | - | 1, 9, 17, 25, 33, 41, 49 |
| 2. Mathematical | - | 2, 10, 18, 26, 34, 42, 50 |
| 3. Musical | - | 3, 11, 19, 27, 35, 43, 51 |
| 4. Spatial | - | 4, 12, 20, 28, 36, 44, 52 |
| 5. Kinesthetic | - | 5, 13, 21, 29, 37, 45, 53 |
| 6. Interpersonal | - | 6, 14, 22, 30, 38, 46, 54 |
| 7. Intrapersonal | - | 7, 15, 23, 31, 39, 47, 55 |
| 8. Naturalist | - | 8, 16, 24, 32, 40, 48, 56 |
| 9. Existentialist | - | 57, 58, 59, 60, 61, 62, 63 |

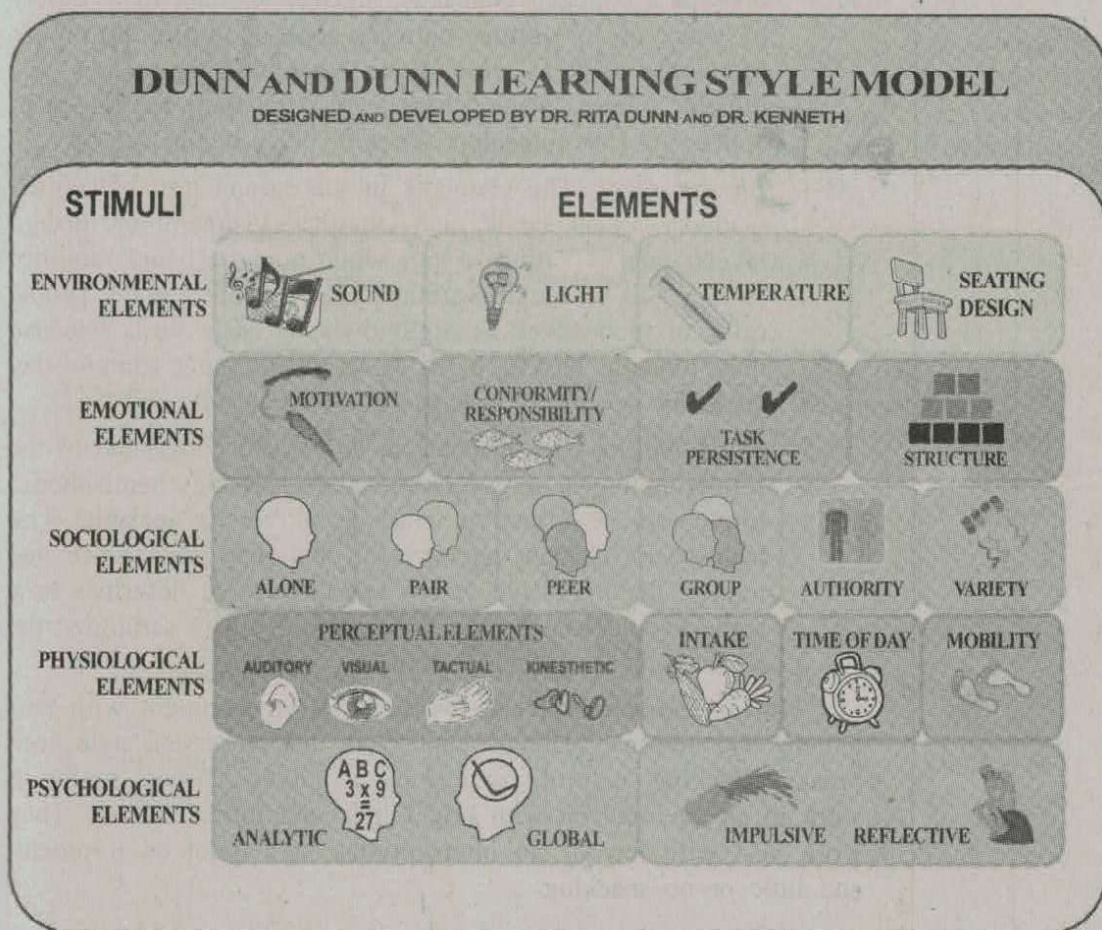
Add your scores per intelligence. Share your dominant intelligence, the top 3 or 4, with your classmates.

All your future students have these multiple intelligences with two or more intelligences more developed than others. If you teach in the same way throughout the year, then you do an act of disservice to your students. To cater to varied multiple intelligences, you must have a variety of teaching methods.

Learning Styles

Another factor that makes your students differ from one another is learning style. Learning style is the way a person processes, internalizes, and studies new and challenging material. Dunn and Dunn present different learning styles according to five (5) groups of stimuli. (Dunn, 2000).

► ► ► **FIGURE 2** Dunn and Dunn Learning Style Model



A summary of these elements is provided below (Dunn, 2000).

1. *Environmental*. The environmental strand refers to these elements: lighting, sound, temperature, and seating arrangement. For example, some people need to study in a cool and quiet room, and others cannot focus unless they have music playing and it is warm (sound and temperature elements).
2. *Emotional*. This strand includes the following elements: motivation, persistence, responsibility, and structure. For example, some people must complete a project before they start a new one, and others work best on multiple tasks at the same time (persistence element).
3. *Sociological*. The sociological strand represents elements related to how individuals learn in association with other people: (a) alone or with peers, (b) an authoritative adult or with a collegial colleague, and (c) learning in a variety of ways or in routine patterns. For example, a number of people need to work alone when tackling a new and difficult subject, while others learn best when working with colleagues (learning alone or with peers element).
4. *Physiological*. The elements in this strand are: perceptual (auditory, visual, tactile, and kinesthetic), time-of-day energy levels, intake (eating or not while studying) and mobility (sitting still or moving around). For example, many people refer to themselves as night owls or early birds because they function best at night or in the morning (time-of-day element).
5. *Psychological*. The elements in this strand correspond to the following types of psychological processing: hemispheric, impulsive or reflective, and global versus analytic. The hemispheric element refers to left and right brain processing modes; the impulsive versus reflective style describes how some people leap before thinking and others scrutinize the situation before moving an inch.

Global learners prefer to work in an environment with soft lighting and informal seating. People with this processing style need breaks, snacking, mobility, and sound. Analytic learners prefer to work in an environment with bright light and formal seating. They work best with few or no interruptions, in a quiet environment, and little or no snacking.



Guide Questions for Discussion

1. The Scholastics formulated this principle: "*Nihil est in intellectu quod non prius fuerit in sensu: There is nothing in the mind which was not first in some manner in the senses.*" What does this mean?
2. Do imperfections in the functioning of the sense organs affect learning? Explain your answer.
3. Can the learner imagine something which he has not first experienced through his senses?
4. In learning, can we do away with memory in our desire to develop higher-order thinking skills in the learners?

5. How do feelings and emotions affect the learner's learning?
6. How different is Gardner's MI theory from the traditional concept of intelligence?
7. Does each student have all of these multiple intelligences? Explain your answer.
8. In the light of Gardner's theory, why does Gardner claim we have been unfair to many of our students in the past as far as teaching and testing strategies are concerned?



Individual Work

1. By means of a concept map, show the faculties or powers of the learner. Be ready to present your work to the class.



For Elaborative Learning

1. If you were to give a symbol for each intelligence and learning style what would you give? Explain each of your symbol.

2. There are nine (9) intelligences and many learning styles given in this Chapter. Does this mean that teacher ought to have twelve different teaching strategies every time? Elaborate on your answer.

3. Do you believe that every child is a potential genius?



Taking it to the Net

1. Read recent researches on memory and imagination. Write at least one recent finding about the power of memory and imagination. Write them here.
2. Research on the following learning styles given by Harvey F. Silver:
 - a. mastery
 - b. interpersonal
 - c. understanding
 - d. self-expressions

These relate to Figure 6 of Unit III, Chapter 3. Figure 6 in Unit III, Chapter 3 will be understood better after reading Harvey F. Silver's Learning Styles.